

# P. SRI SAI CHARAN

AI/ML & Full-Stack Engineering · B.Tech Artificial Intelligence & Machine Learning · Vardhaman College of Engineering  
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## SUMMARY

Second-year B.Tech AI/ML undergraduate with hands-on experience designing end-to-end machine learning systems: anomaly detection pipelines, computer vision classifiers, and REST/WebSocket-backed full-stack applications. Demonstrated problem-solving ability through 105+ LeetCode problems solved (33 Medium, 6 Hard; Contest Rating: 1415). Seeking ML and SWE internship opportunities to contribute to real-world AI/ML systems.

## EDUCATION

### B.Tech in Artificial Intelligence & Machine Learning

Aug 2024 – May 2028

[Vardhaman College of Engineering, Hyderabad](#)

CGPA: 8.45 / 10

**Relevant Coursework:** Data Structures & Algorithms · Machine Learning · Deep Learning · Computer Vision · DBMS · Operating Systems · Software Engineering

## TECHNICAL SKILLS

**Languages:** Python, Java, C, JavaScript, TypeScript, SQL    **AI/ML:** Scikit-learn, TensorFlow, Pandas, NumPy, OpenCV, Isolation Forest, Random Forest, Feature Engineering, Anomaly Detection, Classification, Regression, XAI

**Backend:** Node.js, FastAPI, REST API, WebSockets, Prisma ORM, SQLite    **Frontend:** React, TypeScript, TailwindCSS, HTML5, CSS3

**Tools & Concepts:** Git, GitHub, Jupyter, pytest, System Design, Data Structures & Algorithms

## PROJECTS

### Sports Medicine & Anti-Doping Intelligence Platform · [Full-Stack AI System](#)

Jan 2025 – Present

**Stack:** Python · Isolation Forest · Random Forest · FastAPI · Node.js · WebSockets · React · TypeScript · Prisma ORM · SQLite · OCR

- ▶ Designed and implemented a full-stack biomarker monitoring system: FastAPI REST layer for data ingestion and athlete management, with ML inference results rendered on a React/TypeScript dashboard.
- ▶ Built a two-stage anomaly detection pipeline — Isolation Forest for per-athlete rolling-baseline anomaly detection, followed by a Random Forest classifier assigning low/medium/high risk levels.
- ▶ Integrated Explainable AI (XAI) using Scikit-learn feature importances to surface which biomarkers drove each flagged result inline on the analyst dashboard.
- ▶ Implemented push-based WebSocket alert delivery, pyesseract OCR for lab PDF ingestion, and persistent data layer via Prisma ORM and SQLite.

### AI Forest Fire & Smoke Detection · [Computer Vision Pipeline](#)

2024

**Stack:** Python · OpenCV · Scikit-learn · HSV Color Segmentation · Morphological Filtering

- ▶ Built a multi-stage computer vision pipeline: HSV color-space masking and morphological filtering to isolate candidate regions, then a supervised classifier trained on hand-engineered features with threshold tuning to minimize false negatives.
- ▶ Applied modular software design — separated preprocessing and classification stages for independent component swapping, demonstrating sound ML system architecture.

### AI Athlete Performance Analyzer · [ML Regression](#)

2025

**Stack:** Python · Scikit-learn · Pandas · NumPy · Regression · Feature Engineering

- ▶ Engineered features from raw training logs (load, volume, recovery metrics) and trained regression models to predict near-term athlete performance trends; validated on a held-out test split.

### Shortify — URL Shortening Service · [Backend Engineering](#)

2024

**Stack:** Node.js · REST API · JavaScript

- ▶ Designed a layered backend service for slug generation, collision handling, and redirect resolution, demonstrating production-style system design with clear separation of concerns.

## ACHIEVEMENTS & PROBLEM SOLVING

- ▶ **LeetCode:** 105+ problems solved — 33 Medium, 6 Hard; Contest Rating: 1415; active competitive programming practice.
- ▶ **GitHub:** Multiple public repositories spanning AI/ML and full-stack projects; consistent commit history and documentation.
- ▶ **Portfolio:** Personal technical portfolio deployed at [charan666-one.github.io/portfolio](https://charan666-one.github.io/portfolio).
- ▶ **Academic Standing:** CGPA 8.45/10 in a competitive B.Tech AI/ML program.